

Cyber Tabletop Exercise

Transportation GPS Spoofing Incident

Prepared for: National Logistics Services | Sector: Transportation | Date: 22 June 2026

This document is a facilitation-ready tabletop exercise generated from the provided STIX 2.1 bundle. It is fictional and intended for preparedness, governance, operational resilience, and incident response discussion.

Exercise classification: Internal / exercise use only

1. Introduction

This tabletop exercise simulates a coordinated navigation disruption against National Logistics Services, a transportation organization relying on the Fleet Management Platform for vehicle tracking, route planning, and dispatch operations. The scenario begins with impossible location reports across multiple vehicles and escalates into confirmed GPS signal spoofing that materially affects route optimization, delivery commitments, and national operations.

The exercise is designed for a cross-functional crisis team. It tests whether participants can detect the operational anomaly, distinguish system failure from adversary-driven signal manipulation, safely shift to alternate location-validation procedures, maintain dispatch continuity, and communicate business impact to internal and external stakeholders.

Source STIX Elements

STIX object	Extracted scenario content
Identity	National Logistics Services - transportation sector
Targeted infrastructure	Fleet Management Platform - Platform responsible for vehicle tracking, route planning, and dispatch operations.
Threat actor	Navigation Disruption Actor - type: nation-state
Attack pattern	GPS Signal Spoofing - Manipulation of navigation signals causing incorrect location reporting.
Incident	Fleet GPS Spoofing Incident - GPS spoofing affects vehicle tracking, routing, and delivery operations.

2. Exercise Scope

- Exercise type: discussion-based tabletop exercise.
- Primary scenario: GPS signal spoofing causes inaccurate vehicle locations, automated route disruption, and nationwide service degradation.
- Systems in scope: fleet management platform, dispatch console, vehicle telematics, route optimization, customer delivery communications, crisis management process, and business continuity procedures.
- Systems out of scope: live production testing, offensive technical testing, physical interception of vehicles, and changes to active dispatch operations.
- Estimated duration: 90 minutes, including opening brief, four injects, and debrief.

Assumptions

- Vehicle telematics continue to transmit data, but position data cannot initially be trusted.
- The spoofing effect is geographically dispersed enough to create enterprise-level disruption.
- No confirmed compromise of internal IT systems is known at exercise start.
- Voice communications and manual dispatch procedures remain available.
- Participants should respond according to their actual roles, escalation paths, and authorities.

3. Exercise Objectives

- Validate recognition, triage, and escalation of navigation-data integrity anomalies.
- Clarify when cyber, operational technology, logistics, safety, legal, communications, and executive stakeholders are engaged.
- Test decision-making for switching to alternate navigation sources and manual fleet coordination.
- Assess customer, regulator, partner, and public communications during national delivery impact.
- Identify gaps in fleet telemetry validation, route optimization safeguards, vendor coordination, and crisis governance.
- Capture lessons learned and assign concrete remediation actions.

Audience Focus

Audience	Focus of discussion
Executive / Crisis leadership	Operational risk appetite, service commitments, public messaging, escalation threshold, regulator engagement, and customer prioritization.
Operational / Dispatch	Manual dispatch, driver safety, route validation, delivery prioritization, depot coordination, and continuity of service.
Technical / Cyber / Engineering	Telemetry anomaly detection, spoofing confirmation, isolation of unreliable inputs, validation with alternate sources, vendor coordination, and monitoring.

4. Rules of Engagement

- This is a no-fault learning exercise. The goal is to identify decisions, dependencies, and improvement opportunities.
- Real incidents take priority. Participants may step out if operational needs require it.
- Do not fight the scenario. Assume each inject is accurate at the time presented, even where information is incomplete.
- Use actual organizational processes where possible. Note where the documented process is unclear or unavailable.
- Do not perform live technical actions against production systems as part of the exercise.
- Facilitators may compress or expand the timeline to maintain discussion flow.

5. Scenario Narrative

Navigation Disruption Actor, assessed in the STIX bundle as a nation-state threat actor, targets National Logistics Services using GPS Signal Spoofing. The adversary's objective is not initially data theft or malware deployment, but disruption of trust in navigation data that supports national logistics operations.

The Fleet Management Platform receives location updates that appear valid at first glance, but multiple vehicles begin reporting positions that are physically impossible. Automated route optimization then uses this false location data, causing avoidable reroutes, missed delivery windows, congestion at depots, and increased manual workload for dispatch teams.

As the incident develops, engineering confirms that GPS receivers are accepting spoofed satellite signals. The organization must decide when to distrust automated routing, how to validate vehicle location through alternate sources, how to communicate degraded service, and when to declare a major operational incident.

6. Exercise Timeline and Injects

Inject 1 - Tracking Anomalies | Day 1 - 08:15

Fleet operators observe multiple vehicles simultaneously reporting impossible locations hundreds of kilometers from their expected routes.

Facilitator prompt

Discussion Questions

- What is the first team to own triage: dispatch operations, IT service desk, SOC, or engineering?
- What checks determine whether the issue is a telematics platform failure, data quality defect, or intentional signal manipulation?
- At what threshold should route optimization be paused or placed into advisory mode?
- Who informs drivers and regional depots that location data may be unreliable?

Expected Actions

- Open an operational incident ticket and tag cyber/engineering for anomaly review.

- Compare GPS locations with last known depot scans, driver voice confirmations, cellular triangulation, tachograph/vehicle CAN data if available, and delivery proof-of-presence events.
- Preserve relevant telemetry and dispatch logs for later analysis.
- Advise dispatch not to rely solely on automated location feeds until confidence improves.

Decision point: whether to keep automated routing active while the anomaly is investigated.

Inject 2 - Route Disruption | Day 1 - 10:00

Automated route optimization systems begin rerouting deliveries based on false location data, causing widespread delays.

Facilitator prompt

Discussion Questions

- Which deliveries, routes, customers, or regions should be prioritized while false data is driving delays?
- What criteria justify switching to manual dispatch procedures?
- How are customer-facing delivery estimates updated without overpromising?
- What information should be shared with key suppliers, insurers, regulators, or public-sector transportation partners?

Expected Actions

- Move route optimization to a constrained or manual approval mode.
- Activate business continuity procedures for high-priority deliveries and safety-critical shipments.
- Stand up an incident coordination cadence across operations, cyber, engineering, communications, and legal.
- Track delay metrics and customer impact by region and service tier.

Decision point: whether to declare a major operational incident before technical root cause is confirmed.

Inject 3 - Spoofing Confirmed | Day 1 - 12:30

Engineering teams determine GPS receivers are processing spoofed satellite signals, affecting fleet navigation accuracy.

Facilitator prompt

Discussion Questions

- Who has authority to confirm GPS spoofing as the working cause?
- Which navigation sources can be trusted, and how is trust communicated to dispatchers and drivers?
- Should certain regions or routes be suspended, slowed, or manually escorted through alternate routing?
- What evidence is needed to support law enforcement or government notification?

Expected Actions

- Treat GPS position as untrusted in affected areas and implement alternate validation.
- Use secondary location validation sources and manual dispatch procedures.
- Coordinate with fleet technology vendors and GNSS specialists to validate receiver behavior and mitigation options.
- Prepare executive briefing with operational, safety, customer, and regulatory implications.

Decision point: whether to suspend affected automated delivery operations or continue under manual controls.

Inject 4 - National Delivery Impact | Day 1 - 16:00

Executive leadership declares a major operational incident as nationwide delivery performance drops below service level commitments.

Facilitator prompt

Discussion Questions

- What service-level commitments are now at risk and who approves customer prioritization decisions?
- What is the executive message to customers, regulators, drivers, unions, suppliers, and media?
- What recovery criteria are required before returning to normal automated routing?
- What immediate improvements should be funded after the exercise?

Expected Actions

- Activate crisis management governance and publish a single source of truth for impact reporting.
- Prioritize safety-critical, regulated, perishable, or high-value deliveries.
- Document recovery criteria: verified signal integrity, stable alternate validation, backlog reduction, and stakeholder approval.
- Initiate post-incident review with owners, due dates, and control enhancements.

Decision point: whether to maintain national service under degraded operations, reduce commitments, or temporarily suspend selected routes.

7. Facilitation Guide

Facilitator Role

- Keep discussion focused on decisions, ownership, information needs, and escalation triggers.
- Probe for dependencies between dispatch, cyber, engineering, vendors, communications, legal, and executive teams.
- Capture gaps without solving them during the exercise.
- Ask participants to name the role or function responsible for each action, not just the action itself.

Suggested Timing

Segment	Duration	Purpose
Opening and rules	10 min	Set context, scope, objectives, and rules.
Inject 1	15 min	Initial anomaly triage and escalation.
Inject 2	15 min	Operational continuity and manual dispatch decision.
Inject 3	20 min	Spoofing confirmation and alternate navigation strategy.
Inject 4	15 min	Major incident governance and communications.
Debrief	15 min	Lessons learned, actions, owners, and deadlines.

8. Technical and Business Impact Areas

Technical Impacts

- Loss of confidence in GPS-derived location reporting.
- Route optimization decisions based on false or corrupted inputs.
- Potential alert fatigue from simultaneous vehicle anomalies.

- Need to preserve telemetry, receiver data, dispatch logs, and engineering findings.
- Increased reliance on manual procedures and secondary validation sources.

Business and Operational Impacts

- Missed delivery windows and breach of service-level commitments.
- Backlog accumulation at depots and customer sites.
- Safety concerns if drivers follow incorrect navigation data.
- Customer complaints, reputational exposure, and possible contractual penalties.
- Executive decision pressure around prioritization, disclosure, and degraded service.

Control Considerations

- Switch to Alternate Navigation Sources: Use secondary location validation sources and manual dispatch procedures.
- Manual Fleet Coordination: Coordinate vehicle movements using voice communications and validated location reporting.
- Implement anomaly thresholds for impossible movement, sudden jumps, and inconsistent multi-source location signals.
- Design dispatch workflows that allow automated routing to be placed into advisory-only mode.
- Maintain an approved communications playbook for large-scale delivery disruption.
- Establish evidence retention requirements for external signal manipulation events.

9. Debrief and Lessons Learned

Use the following questions to close the exercise and convert observations into improvement actions.

- What information was missing in the first 30 minutes, and who should have provided it?
- Were escalation thresholds clear between dispatch, IT, cyber, engineering, and crisis leadership?
- Did the organization have a documented process to distrust or override automated route optimization?
- Were alternate navigation and manual dispatch procedures realistic for a nationwide event?
- Were customer, regulator, partner, and public communications coordinated through a single approved channel?
- What controls or playbooks should be updated within 30, 60, and 90 days?

Improvement item	Owner	Due date	Status
1.			
2.			
3.			
4.			
5.			
6.			
7.			

10. MITRE ATT&CK Mapping

This scenario maps most directly to MITRE ATT&CK for ICS because the primary effect is manipulation of operational telemetry and process control decisions rather than conventional enterprise endpoint compromise. The mapping below is intentionally pragmatic and should be validated against the organization's telemetry architecture.

Tactic	Technique / ID	Scenario relevance
Evasion / Impair Process Control	Unauthorized Message: Reporting Message - T1692.002	Spoofed GPS/location data functions like false operational telemetry: receivers and platforms process reporting data that does not reflect real vehicle location.
Impair Process Control	Manipulation of Control - T0831	False navigation inputs influence automated route optimization and dispatch decisions, impairing the process the fleet platform is meant to control.
Impact	Loss of Productivity and Revenue - T0828	Nationwide delivery delays and reduced service-level performance create business disruption and revenue exposure.
Impact	Loss of View - T0829	Dispatchers lose reliable visibility into where vehicles actually are, even though the platform continues to display location data.
Impact	Network Denial of Service - T1464 (Mobile, related)	Related consideration where GNSS jamming, rather than spoofing, degrades GPS availability. Included as adjacent detection and response context, not the core scenario technique.

References: MITRE ATT&CK for ICS Matrix and Techniques; MITRE ATT&CK Unauthorized Message: Reporting Message; MITRE ATT&CK Manipulation of Control; MITRE ATT&CK Mobile Network Denial of Service.

Appendix A - Glossary

Term	Definition
GPS / GNSS	Satellite-based navigation systems used to determine position, timing, and movement.
GPS spoofing	Transmission or manipulation of navigation signals so receivers calculate an incorrect position or time.
Telematics	Vehicle data collection and transmission, commonly including location, movement, diagnostics, and driver events.
Route optimization	Automated routing logic that uses location, capacity, time windows, traffic, and priority data to plan delivery movement.
Manual dispatch	Human-led coordination of vehicle movements using verified location reporting, voice communications, depot updates, and controlled routing decisions.
SOC	Security Operations Center responsible for cyber monitoring, triage, and incident support.